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Research Statement

LLM Fine-Tuning for Mechanics

My initial research explored adapting a general-purpose language model to question answering in mechanics of materials. I examined existing instruction datasets and prepared question–answer pairs and concept explanations from textbooks and technical literature to construct domain-specific training data. Using these data, I fine-tuned Qwen2.5-7B with low-rank adaptation (LoRA). I released two instruction datasets and two model checkpoints on Hugging Face, making the training resources and fine-tuned models publicly available.

ConvLSTM Prediction of Battery Material Fields

I studied autoregressive prediction of lithium-concentration and von Mises stress fields in battery materials using conditional ConvLSTM models. I automated MATLAB–COMSOL simulations to construct a dataset in which each sample contained 25 time-aligned frames of each field, a grain-orientation map, and a C-rate map. I trained separate models in PyTorch for the two physical fields. At every prediction step, each model jointly received five consecutive frames of its target field and the two condition maps to predict the next frame. When predictions were fed back into the five-frame input window, errors accumulated rapidly and the predicted fields became distorted. I therefore introduced scheduled sampling, gradually incorporating model-generated frames into the input windows during training. Comparing autoregressive predictions against COMSOL reference sequences, I found that this approach reduced errors over the same prediction horizon and delayed visible distortion relative to training without scheduled sampling.

PINNs for Shock Capturing

I next studied physics-informed neural networks (PINNs) for shock propagation in the one-dimensional inviscid Burgers equation. Using PyTorch and automatic differentiation, I implemented a standard PINN with a training loss incorporating the PDE residual, initial conditions, and boundary conditions. I then added an artificial-viscosity term with a fixed coefficient to the PDE residual and compared the resulting model with the standard PINN, using the analytical inviscid solution as a reference. In this comparison, the artificial-viscosity model achieved a lower relative L^2 error but a larger maximum pointwise error. I examined the shock location and transition width alongside these metrics to assess how the two formulations represented the discontinuity. This comparison showed that an improvement in aggregate error did not necessarily indicate a more accurate local representation of the shock.

Hybrid Finite Element–Neural Operator Solvers

During my research internship at Johns Hopkins University, I began working on a hybrid finite element–neural operator solver under the supervision of Professor Somdatta Goswami.

The framework uses non-overlapping domain decomposition, with finite element and neural-operator subdomains exchanging information across a shared interface. I use GPU-accelerated JAX-FEM simulations to generate paired training data and train Transolver in JAX to learn the mapping from interface displacement fields to subdomain stress fields. During coupling, the predicted stresses are used to compute interface reaction forces for the finite element solver. The finite element solver then updates the interface displacements, which are passed back to the neural operator to predict the subdomain stress fields in the next coupling iteration. This coupling process helped me recognize that the operator's predictions influence the interface conditions it subsequently receives. I therefore examine stress-prediction accuracy together with interface displacement compatibility, force balance, and convergence of the coupled solver. I also distinguish the speed of a single neural-operator prediction from the total computational cost of obtaining a sufficiently accurate coupled solution.

The neural operator initially performed well on a validation set drawn from the Gaussian random field (GRF) dataset, but its stress predictions were much less accurate for interface displacements encountered during coupling. I identified a mismatch between the sampled displacement fields and those arising in coupling calculations. Using displacement fields from coupling calculations as a reference, I adjusted the GRF length scale and amplitude to generate new interface displacement samples. I then used JAX-FEM to compute the corresponding subdomain stress fields and retrained Transolver on the revised paired dataset. Tests within the FEM–NO solver for both cylindrical and bracket geometries showed lower stress-prediction errors for coupling-generated inputs, while performance on the GRF validation set remained good. This comparison showed that validation within the generated dataset did not fully represent the operator's performance in the coupled solver. I also explored convergence acceleration strategies and relaxation parameters. I am currently extending this work to three-dimensional fiber-reinforced composites, with neural-operator weights shared across fiber subdomains.

Collaboration and Open Research

During the China International College Students' Innovation Competition, I worked with my teammates on language-model fine-tuning. We exchanged ideas, discussed unfamiliar tools, and worked through technical questions together, ultimately earning an award as a team. I also released instruction datasets and model checkpoints on Hugging Face and documented my research projects in public repositories. In future work, I hope to continue sharing research resources and explaining my methods clearly, so that others can understand how the work was carried out and build on it.

Future Work

Working with JAX-FEM and training Transolver in JAX has strengthened my interest in differentiable programming and simulation. I would like to explore how neural operators and numerical solvers can form an end-to-end differentiable computational graph, connecting material parameters, boundary conditions, or design variables to physical responses and objective functions. Automatic differentiation can then compute gradients of an objective function with respect to these inputs, supporting parameter identification, design optimization, and PDE control. I am interested in how neural-operator prediction errors affect these gradients as the inputs change during these tasks. In gradient-based optimization, the computed gradients guide updates to material parameters or design variables. I want to understand whether a neural operator trained to predict physical responses also captures how those responses change with these variables. Accurate response predictions alone do not guarantee accurate gradients, and errors in these gradients could lead an optimization in an unhelpful direction. This raises a question I would like to investigate: whether training on response values alone is sufficient for these applications, and how the training data or loss function might need to change when the operator is used to guide optimization.

Through doctoral study, I hope to become an independent researcher who can formulate meaningful questions and take responsibility for pursuing them. I want to develop the judgment to explain why I choose an approach and what I can reasonably conclude from its results. Working with teammates and researchers from different backgrounds has shown me how discussion can reveal gaps in my understanding and lead me to reconsider an idea. I hope to contribute my own questions and experience to these exchanges, while remaining willing to revise my thinking in response to evidence and feedback. In the longer term, I would like to pursue research in scientific computing, including opportunities in industrial research environments such as NVIDIA. Wherever I work, I want to combine the initiative to develop my own ideas with the openness to learn from others and work through difficult questions together.